

The Wang Method Series
Notebook #7 (Version 1.0)
Comparative Study and Educational Perspectives

Laung-Terng (LT) Wang

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1. Motivation

The last two centers (L2C) of the 5×5 cube can be solved using different strategies. Rather than viewing them as competing methods, they should be regarded as different optimizations for different types of cubers.

This section compares three representative approaches:

1. Noah Richardson's Beginner Method
2. CubeSkills Method
3. The Method Presented in This Notebook

Each method emphasizes a different design objective. Figure 7.1 illustrates the different L2C structures constructed by the three methods: a cross on each of the top and front faces, two vertical columns on the front face, and two vertical columns on each of the top and front faces.

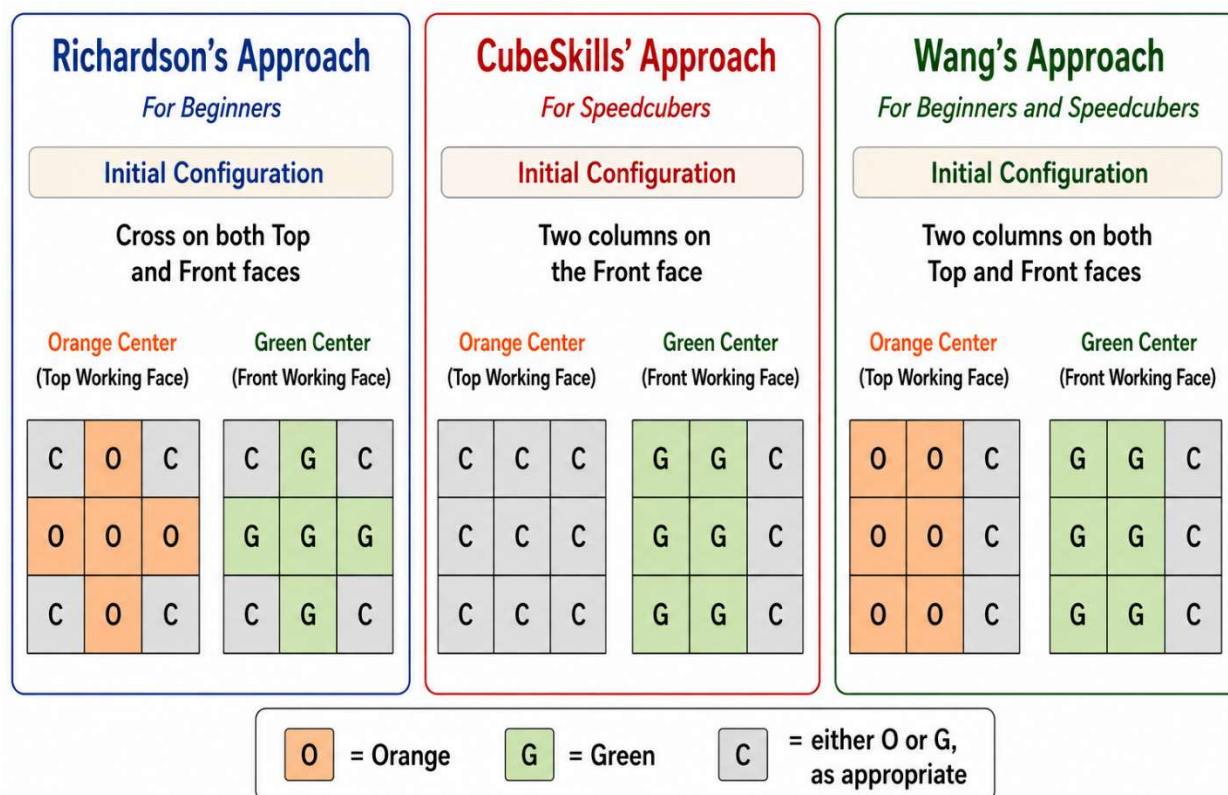


Figure 7.1. Initial configurations used by three common approaches for building the first structures when solving the last two centers on the 5x5 Rubik's Cube.

2. Noah Richardson's Beginner Method

Noah Richardson's method is specifically designed for beginners [1]. Rather than recognizing many different L2C cases and memorizing numerous algorithms, the method follows a fixed construction procedure.

First, a cross is built on one center. A cross is then built on the opposite center. After the two crosses have been completed, only the four corner positions on each center remain unsolved.

Unlike methods that recognize many representative patterns, Noah's method repeatedly applies a single primitive algorithm (Right Sune), together with simple cube rotations, until all remaining corner pieces are solved. This greatly reduces the memorization burden and makes the method particularly attractive for beginning cubers.

Remaining Corner Configurations

After both crosses have been completed, only the corner pieces remain unsolved.

There are

- four corner positions on the orange center, and
- four corner positions on the green center.

Therefore, eight corner positions remain.

Among these eight positions are exactly

- four orange center pieces, and
- four green center pieces.

Since pieces of the same color are indistinguishable, a complete corner configuration is determined simply by selecting which four of the eight positions contain the orange pieces. The remaining four positions must then contain the green pieces.

Hence, the total number of possible corner configurations is

$$\binom{8}{4} = \frac{8!}{4! 4!} = 70$$

This simple combinatorial argument provides an intuitive explanation for why exactly 70 corner configurations are possible after the two crosses have been completed.

Mismatched Corner Pairs

Suppose exactly k orange pieces are located on the green center.

Since the numbers of orange and green pieces are equal, there must also be exactly k green pieces located on the orange center.

Each misplaced orange piece together with its corresponding misplaced green piece forms one **mismatched corner pair**.

To obtain exactly k mismatched corner pairs,

- choose k of the four green-center corners to contain orange pieces, and
- choose k of the four orange-center corners to contain green pieces.

Therefore,

$$N(k) = \binom{4}{k} \binom{4}{k} = \binom{4}{k}^2$$

and the corresponding probability is

$$P(k) = \frac{\binom{4}{k}^2}{\binom{8}{4}} = \frac{\binom{4}{k}^2}{70}$$

Solving the Remaining Corner Pairs

The key idea of Noah Richardson's beginner method is remarkably simple.

One application of the 7-move Right Sune algorithm, together with appropriate setup rotations, swaps and corrects **one mismatched corner pair**.

Consequently,

- one mismatched pair requires one Right Sune;
- two mismatched pairs require two Right Sunes;
- three mismatched pairs require three Right Sunes;
- four mismatched pairs require four Right Sunes.

Therefore, if k mismatched corner pairs remain,

$$M(k) = 7k$$

where $M(k)$ denotes the required move count after the two crosses have been completed.

The complete probability distribution is summarized in Table 7.1.

Table 7.1. Probability distribution of Noah Richardson's Beginner Method.

Mismatched Corner Pairs	Configurations	State Distribution	Right Sune Applications	Moves
0	1	1.43%	0	0
1	16	22.86%	1	7
2	36	51.43%	2	14
3	16	22.86%	3	21
4	1	1.43%	4	28
Total	70	100%	—	—

The distribution

$$1 : 16 : 36 : 16 : 1$$

is perfectly symmetric. The most common situation consists of two mismatched corner pairs, accounting for over half of all possible configurations.

Average Move Count

The theoretical average move count is obtained by taking the probability-weighted average:

$$E[M] = \frac{1(0) + 16(7) + 36(14) + 16(21) + 1(28)}{70} = \frac{980}{70} = 14$$

Equivalently, the expected number of mismatched corner pairs is

$$E[k] = \frac{0(1) + 1(16) + 2(36) + 3(16) + 4(1)}{70} = \frac{140}{70} = 2$$

and since each mismatched corner pair requires one 7-move Right Sune,

$$E[M] = 7E[k] = 7 \times 2 = 14$$

Thus, after the two crosses have been completed, Noah Richardson's beginner method requires an average of **14 moves** to solve the remaining corner pairs.

The principal strength of Noah Richardson's method lies in its simplicity rather than move efficiency.

By repeatedly applying the same primitive algorithm, beginners need to memorize only one core transformation while following a fixed construction strategy. Although the move count may vary from 0 to 28 moves depending on the number of mismatched corner pairs, the theoretical average is exactly **14 moves**.

This elegant trade-off between memorization and move count makes the method an excellent educational approach for beginning cubers.

3. CubeSkills Method

The CubeSkills approach targets speedcubers rather than beginners [2]. Before applying its L2C algorithms, the solver first constructs a solved 2×3 center block on the front face.

CubeSkills states:

These L2C algorithms are for the cases where a 2×3 block of center pieces is solved on the front face. After this construction, CubeSkills presents **20 representative patterns**, each associated with a direct algorithm.

The listed solutions typically require approximately

- 3 moves,
- 7 moves,
- or 11 moves,

depending on the configuration.

To assist memorization, CubeSkills places parentheses around portions of longer algorithms, grouping familiar move triggers into smaller units.

This organization is particularly useful for experienced speedcubers who are already accustomed to memorizing large collections of algorithms such as one-look OLL and one-look PLL.

For such cubers, memorizing an additional set of L2C algorithms may represent a reasonable trade-off for shorter solutions.

Completeness

CubeSkills presents 20 practical patterns.

However, the reference does not appear to provide a mathematical proof that these 20 patterns constitute the complete set of all reachable L2C configurations after the prescribed 2×3 construction.

Consequently, the 20 CubeSkills patterns should be viewed as an effective speed-solving algorithm library rather than a mathematically derived classification of the complete state space.

4. The Method Presented in This Notebook

Unlike the previous two approaches, this notebook emphasizes understanding of the underlying state space [3]. The analysis begins by reducing the remaining configurations into canonical representative states through symmetry, rotations, and swap operations.

This framework naturally supports two operating modes.

Beginner Mode

The beginner mode minimizes memorization. Representative cases are normalized by swap and rotation so that only TT and MM algorithms are required.

The resulting solutions typically require approximately 11-12 moves, while requiring only a very small algorithm library. This mode is intended for casual cubers who solve cubes only occasionally.

Direct Mode

The direct mode eliminates normalization. Instead, each representative case is solved directly using its own algorithm.

For the representative cases currently identified, the solution lengths are approximately

- 4 moves,
- 8 moves,
- 11 moves,

including one setup rotation when required.

Compared with the beginner mode, this approach requires memorizing additional representative algorithms but consistently produces substantially shorter solutions. Twelve representative algorithms are sufficient.

5. Overall Comparison

Table 7.2 compares key features of the three methods. Beginners and speedcubers can therefore choose the approach best suited to their needs.

Table 7.2. Comparison of key features of three L2C solving approaches.

Feature	Noah	CubeSkills	This Notebook
Primary Objective	Minimize memorization	Minimize move count	Balance memory and move count
Target Users	Beginners	Speedcubers	Beginners to advanced cubers
Initial Construction	Two crosses	2×3 front block	Canonical-state reduction
Representative Cases	Repeated primitive algorithm	20 listed patterns	Canonical representative states
Algorithms	One core algorithm	About 20 direct algorithms	TT/MM mode or Direct mode
Typical Moves	Average 14	3–7–11	Beginner: 11–12; Direct: 4–8–11
Main Strength	Simplicity	Speed	Flexibility and state-space understanding

A More Flexible Approach

In practice, cubers do not necessarily need to follow any one of the three approaches described above. Another simple strategy is to place as many center pieces as possible on their correct Top and Front faces without following a predetermined construction pattern. Once only a few opposite-color pieces remain on the two centers, they can be solved one pair at a time using familiar TT or MM commutators. This provides a flexible alternative for cubers who prefer intuitive piece placement over a fixed center-building structure.

6. Discussion

The three methods presented in this chapter should not be viewed as competing approaches. Instead, they represent three different design philosophies for solving the last two centers (L2C), each optimizing a different objective.

Noah Richardson's method simplifies the **construction process**. By building two crosses first, the remaining configurations are greatly simplified and can be solved by repeatedly applying a single primitive algorithm (Right Sune). This elegant construction-based strategy minimizes algorithm memorization and provides an excellent learning path for beginners.

The CubeSkills method simplifies the **solution length**. After constructing a 2×3 center block, it recognizes representative patterns and applies direct algorithms, typically requiring only 3, 7, or 11 moves. This approach is well suited for competitive speedcubers who are willing to memorize a larger collection of algorithms. However, the CubeSkills reference presents 20 practical patterns without establishing mathematically whether they constitute a complete classification of all reachable L2C configurations under its construction.

The method presented in this notebook simplifies the **state-space analysis**. Through canonical-state reduction, symmetry, rotations, and swap operations, the remaining configurations are organized into a systematic analytical framework. This framework supports two operating modes. The beginner mode minimizes memorization through normalization, whereas the direct mode trades additional memorization for consistently shorter solutions.

Readers may therefore select the approach that best matches their own objectives:

- **Beginners** may prefer Noah Richardson's simple construction method.
- **Casual cubers** may prefer the notebook's normalized TT/MM beginner mode with its small algorithm library.
- **Intermediate and advanced cubers** may choose the notebook's direct representative algorithms for shorter solutions.
- **Competitive speedcubers** may adopt larger algorithm libraries such as CubeSkills to minimize move count.

Ultimately, these three methods complement one another rather than compete with one another. Together they illustrate three different perspectives on solving the last two centers:

- **Noah Richardson's method** emphasizes construction.
- **CubeSkills** emphasizes move efficiency.
- **This notebook** emphasizes mathematical understanding and state-space analysis.

Each perspective offers valuable insights into the L2C problem. Taken together, they demonstrate how different optimization objectives—construction strategy, algorithm memorization, move efficiency, and mathematical analysis—can lead to different, yet equally elegant, solutions.

References

- [1] Noah Richardson, “ 5×5 LAST 2 CENTERS | EASY TUTORIAL,” YouTube.
<https://www.youtube.com/watch?v=Vv5j1TyZ6Ew>
- [2] CubeSkills, “ 5×5 L2C Algorithms (Last Two Centers),” *Last 2 Centers Algorithms [5x5]*.
<https://cubeskills.com/uploads/pdf/tutorials/last-2-centers-algorithms-5x5.pdf>
- [3] L.-T. Wang, *Understanding the Last Two Centers of the 5×5 Rubik's Cube: A Memory-Oriented Framework*, The Wang Method Series, Notebook #6, Version 1.0, Aug. 15, 2026.
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